

6. TESTING

6.1 Acceptance Testing

Acceptance tests ensure that the system satisfies the requirements set at the beginning of the project. The objective is to verify that the system delivers what the stakeholders desire before its delivery or usage.

The acceptance criteria are as follows:

- The users should be able to log into the system and get an access token.
- The authorized user should add a new animal and retrieve it using the API.
- The CSV file containing movement information should be able to upload and saved.
- The search filters should return the right animals according to species or geographical position.
- API should return proper HTTP error messages (401, 400, 404).
- The admin should be able to create a new user with the role.

6.2 Test Cases

TC ID	Feature	Input	Expected Output
TC-01	User Login	Valid email and password	JWT token returned, status 200
TC-02	User Login (wrong)	Valid email, wrong password	Status 401 Unauthorized
TC-03	Add Animal	Valid animal data fields	Record saved, new ID returned
TC-04	Add Animal (missing)	Species name missing	Status 400, field error message
TC-05	Get Animal by ID	Valid animal ID in URL	Full animal record in JSON
TC-06	Get Animal (wrong ID)	ID that does not exist	Status 404 Not Found
TC-07	Search by Species	species=Snow Leopard query	List of matching animals
TC-08	CSV Upload (valid)	Correctly formatted CSV file	All rows saved successfully
TC-09	CSV Upload (bad)	CSV with missing coordinates	Rows skipped, error report shown
TC-10	API without a token	GET request without Auth header	Status 401 Unauthorized
TC-11	Export Data	Request to export search results	JSON or CSV file downloaded
TC-12	Admin creates user	New user details with role	User account created successfully

6.3 Test Plan

Testing Approach: Black Box Testing

The black box testing approach will be used as the main testing technique by the development team.

Tests are created solely based on input values and expected output results.

Black box testing is appropriate for this case due to the following reasons:

- There are specific inputs and expected outputs of API endpoints.
- Functional requirements will be validated without having to read the code.
- This reflects the way actual users and other applications will work with the product.

Test Levels

Unit Testing: Unit tests will be conducted for each API endpoint with the help of Postman. In each unit test, only one API endpoint is tested.

Integration Testing: Integration testing will be carried out on the completion of development of all the API endpoints, and they will be tested together.

System Testing: System testing will involve conducting tests for the entire application system from logging in to exporting the data via APIs.

Acceptance Testing: The final stage of testing that involves testing against the requirements specification.

Test Tools

Postman: For manual API testing.

Browser DevTools: For testing the web interface.

Test Schedule

Phase	Test Type	When
After each endpoint	Unit Test (Postman)	During development
After full API build	Integration Test	After Increment 2
Before submission	System & Acceptance Test	Final week

Definitions & Acronyms

Term / Acronym	Definition
WIS	Wildlife Information System
API	Application Programmable Interface
REST	Representational State Transfer
JWT	JSON Web Token

FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
DFD	Data Flow Diagram
ER	Entity-Relationship